Chaotic Fractal Based RNG Algorithm

K Dhiraj
School of Computer Science and Engineering (SCOPE)
Vellore Institute of Technology
Chennai, India
dhiraj.k2023@vitstudent.ac.in

M Vaishnavi
School of Computer Science and Engineering (SCOPE)
Vellore Institute of Technology
Chennai, India
vaishnavi.m2023@vitstudent.ac.in

Dr. Srisakthi Saravanan
School of Computer Science and Engineering (SCOPE)
Vellore Institute of Technology
Chennai, India
srisakthi.saravanan@vit.ac.in

Corresponding Author: Dr. Srisakthi Saravanan, Email ID srisakthi.saravanan@vit.ac.in

***Abstract*—Random number generation plays a vital role in cryptography, simulations, stochastic modelling, and secure communications. Traditional PRNGs usually adopt arithmetic recurrences that may be subject to predictability and limited entropy. In this paper, we present a new Chaotic Fractal scale RNG algorithm which explores the polygonal geometry, point-in-polygon dynamics, and nonlinear iterative mappings. Due to deterministic chaos, the algorithm shows high sensitivity in initial conditions and displays formidable statistical unpredictability. Experimental evaluation shows that the proposed approach provides a significantly higher computational efficiency compared to traditional spatial random point generators while preserving all the desirable randomness properties.**

***Index Terms*—Random Number Generation (RNG), Pseudo-Random Numbers (PRNs), True Random Numbers (TRNs), Fractal Geometry, Chaotic Systems, Geometric Boundary Constraints, NIST SP-800-22 Test, NIST SP 800-90B**

I. Introduction

Randomness becomes a basic requirement for modern computing, from cryptographic key generation to Monte Carlo simulations. Although hardware-based true random number generators give a very good level of entropy, they often come at high prices and depend on the host platform. Therefore, the use of software-based PRNGs is still thriving and thus induces inherited limitations, such as periodicity and susceptibility to reverse engineering. Chaotic systems are deterministic, yet extremely sensitive to initial conditions, and for that reason exhibit behaviour apparently random in the long run. Chaotic fractals can be a good candidate, since they can generate randomness easily from an iterative process of fractal structures. In this paper, we propose an RNG which is based on polygon fractal generated by chaotic iteration inside the polygon and using the spatial evolution of points to derive random values. Before discussing our methodology, it is essential to understand the concepts of fractal structures and chaotic algorithms. The concept of fractals was introduced by Benoît B. Mandelbrot in his work The Fractal Geometry of Nature [4], where he demonstrated that many natural phenomena, such as coastlines, clouds, and mountain ranges, exhibit self-similar patterns at different scales. Mandelbrot also explored chaotic systems, showing how irregular, self-replicating structures emerge from chaos. Over time, researchers leveraged fractals to develop chaotic algorithms for random number generation, which have been applied in various fields, including image encryption and data security [6] [7]. However, chaos-based random number generators (RNGs) come with inherent challenges:

A. Computational Overhead

Many chaotic encryption algorithms involve complex mathematical operations, increasing computational costs [2] [3].

B. Deterministic Nature

Despite their random appearance, chaotic systems are inherently deterministic, meaning they can be reconstructed if the parameters are known [5].

C. Difficulty in Passing Statistical Tests

RNGs based on chaotic systems sometimes fail established randomness tests, such as the NIST suite or Diehard tests, making them unsuitable for certain high-security applications [8].

In this paper, we propose an algorithm that addresses these challenges. By utilizing basic mathematical functions, we aim to reduce computational overhead while introducing specific variables to ensure that backtracking and predicting subsequent numbers remain nearly impossible, thereby enhancing the overall randomness and unpredictability of the generated sequence.

This work explores a chaotic fractal-based random number generation approach grounded in polygonal geometry. The algorithm operates within bounded polygonal domains and iteratively generates point trajectories through nonlinear geometric transformations. A key motivation of this study is to investigate how polygonal dimensionality influences randomness generation. To this end, the proposed method is evaluated across multiple polygon dimensions, allowing geometric complexity to act as a controllable source of entropy. As the number of polygon vertices increases, the resulting chaotic dynamics become more intricate, leading to enhanced unpredictability in the generated sequences. By integrating geometric constraints with iterative chaotic mappings, the proposed approach introduces an additional degree of freedom absent in traditional pseudo-random number generators. The

ability to tune randomness characteristics through polygon dimension, without altering the underlying algorithmic structure, positions this method as a flexible and efficient alternative for applications requiring high-quality random number generation.

The motivations are two-fold: (i) leveraging geometric chaos for randomness generation, and (ii) constructing an algorithm that is computationally efficient with potential use in large-scale random number generation.

II. Literature Review

Chaotic, fractal-based random number generation has emerged as an important research issue in cryptography and secure communications because they are able to generate sequences with high randomness and unpredictability [10], [14]. These algorithms are well suited for cryptographic applications due to the fact that they have key properties of chaotic systems such as sensitivity to initial conditions, topological mixing and ergodicity. The addition of fractal geometry further increases the complexity and unpredictability of the generated sequences [11] [14].The use of principles of fractal geometry tends to further increase the complexity and unpredictability of these generators [10] [15].

The main concept behind chaotic pseudo-random number generators (PRNGs) is to utilize deterministic chaotic maps mathematical functions that exhibit complex, non - repetitive results even after being governed by simple rules [13] [14]. Logistic map, tent map, various 2D and 3D hyperchaotic systems are some of the commonly used chaotic maps [9] [11]. For example the logistic map demonstrates how complex and unpredictable behaviours like chaos can be obtained from deterministic rules, undergoing a period doubling cascade to chaos as its parameter increases [16]. SSimilarly, the tent map is appealing for generating pseudorandom sequences. This is because it is easy to implement and does not have stability islands in its chaotic regime. [23].

The generation of pseudorandom numbers by using chaotic maps involves iterating these maps, with the output at each step adding to the random sequence. Because they are extremely sensitive to initial conditions and control parameters, their output sequences are unpredictable; a small change in these values can result in quite different output sequences [17] [18] [12] [13]. This thing is, like the "butterfly effect" that happens in complicated systems. The butterfly effect is when small things that happen at the start make a difference later on. These small differences get bigger and bigger making long-term predictions impossible. [9].A positive Lyapunov exponent acts as a definitive indication of a chaotic system, providing a quantitative measure of the exponential divergence [9].

Individual chaotic maps, like the logistic map or tent map, can make random-looking sequences. However, they usually struggle with problems like limited chaotic regions and tiny keyspaces. Plus, their randomness can get worse in computers because of precision issues [11] [13] [16] [22]. Researchers have proposed several strategies for addressing this type of problem; one of the most commonly used is to combine or connect more than one chaotic map to produce a single chaotic map. Understanding these different types of maps has provided a greater degree of robustness through the generation of a wider range of random sequences [17]. For example, combining the logistic map and the tent map, especially with techniques like deep-zoom analysis, has been found to improve pseudorandom qualities [17] [18]. Other possible hybrid approaches include combination of nonlinear shift registers with chaotic maps [16].

The idea of 'robust chaos' is also of great importance; it is a concept defining the chaotic system, that have robust chaotic characteristics over larger parameter space with the positive Lyapunov exponent. Robust chaotic systems are best suited for designing cryptographically secure PRNG (CSPRNG) [12] [22]. Robust Chaotic Tent Map (RCTM) proposed for CSPRNG design is one of the example for these [12] [22]. Furthermore, novel 2D hyperchaotic maps, such as those designed either from 1D chaotic maps or by combining the Hénon map and Sine map, have been developed to overcome issues like low uniformity and randomness in simpler chaotic systems [11] [19].

Chaotic systems generate patterns that appear at different scales throughout history, from the Matryoshka dolls of Russian fairy tales and Victor Vasarely's optical paintings to the classic examples of fractal geometry in nature - like the coast of Cornwall - which are formally defined as zero-dimensional sets with infinite detail [9] [10].These intricate patterns are a direct consequence of the stretching and folding mechanisms inherent in chaotic dynamics [9]. The fractal nature of these systems can be used to design PRNGs, where the generation process often involves trajectories evolving on these fractal patterns[10]. Additionally, the term "fractal function" has been used to describe a novel nonlinear pseudorandom sequence generator [15].

Table I summarizes the key fractal and chaos based random number generation approaches discussed in the literature, highlighting the techniques used by the authors and their primary contributions in the study.

III. Proposed Chaootic Fractal RNG Algorithm

The proposed algorithm operates within an arbitrary degree of simple polygon. Starting from an initial point strictly inside the polygon and an initial control parameter V, the algorithm generates a sequence of points using a nonlinear deterministic rule. Every iteration yield a new interior point of the polygon and an updated control parameter. The trajectory of the points creates a fractal-like structure because of the chaotic nature of the mapping.

We have broadly categorized all the operations in our algorithm into five key stages, as illustrated in Figure 1. In the following sections, we will examine each step individually and analyse its implementation in detail.

A. Regular Polygon Initialization

Algorithm 1 presents a deterministic procedure for constructing a regular n-sided polygon in the Euclidean plane given a fixed edge length. The algorithm begins by computing

TABLE I
COMPARISON OF FRACTAL AND CHAOS BASED RNG APPROACHES

Ref	Technique	Category	Contribution
[5]	Fractal based number generator	Fractal	Designed a fractal based random number generator.
[12]	Robust chaotic tent map (RCTM)	Chaotic	Secure PRNG based on robust chaotic mapping for cryptography .
[13]	Enhanced logistic mapping	Chaotic	Improved cryptographic security in random number generation process.
[17]	Logistic map + Tent map + Deep zoom	Hybrid Chaotic	Improved pseudorandom properties in random number generation through map composition and deep-zoom analysis.
[18]	2D discrete and hyperchaotic System	Hyperchaotic	Worked on high speed pseudorandom bit generation using hyperchaotic systems.
[19]	2D Henon–Sine hyperchaotic map	Hyperchaotic	Novel PRNG with microcontroller implementation.
[20]	Piecewise logistic map	Chaotic	PRNG based on piecewise logistic map.
[22]	Modified skew tent map	Chaotic	Application of modified skew tent map in desiging PRNG .
[23]	Sine function + Skew tent map composition	Hybrid Chaotic	Function and composition based pseudorandom number generation.
[15]	Fractal function	Fractal	Novel nonlinear pseudorandom sequence generator using fractal functions.

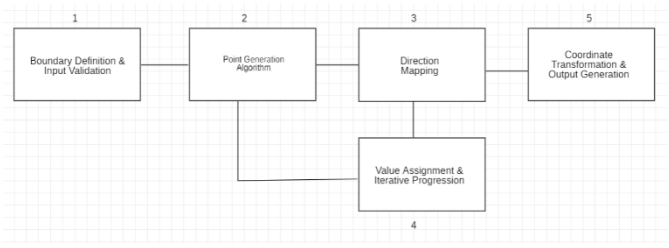


Fig. 1. Methodology Flowchart of the Proposed System

Algorithm 1 Regular Polygon Initialization

Require: Edge length $a > 0$, number of sides $n \geq 3$
Ensure: Ordered vertex set $\mathcal{P} = \{v_1, v_2, \ldots, v_n\} \subset \mathbb{R}^2$

Step 1: Compute Circumradius
$R \leftarrow \frac{a}{2\sin(\pi/n)}$
Step 2: Compute Center Coordinates
$x_c \leftarrow \frac{a}{2\tan(\pi/n)}$
$y_c \leftarrow \frac{a}{2}$
Step 3: Compute Initial Angle
$\theta_0 \leftarrow \mathrm{atan2}(-y_c, -x_c)$
Step 4: Generate Vertices
$\mathcal{P} \leftarrow \emptyset$
for $k = 0$ **to** $n - 1$ **do**
  $\theta_k \leftarrow \theta_0 + k\left(\frac{2\pi}{n}\right)$
  $v_{k+1} \leftarrow (x_c + R\cos(\theta_k), y_c + R\sin(\theta_k))$
  $\mathcal{P} \leftarrow \mathcal{P} \cup \{v_{k+1}\}$
end for
return $\mathcal{P}$

the circumradius of the polygon using standard geometric relationships for regular polygons, ensuring that all vertices lie on a common circumscribed circle. A reference center point is then defined to position the polygon within a set of real numbers. To guarantee a consistent orientation independent of translation, an initial reference angle is computed using the atan2 function. This choice provides a robust angular reference that remains valid across all quadrants of the coordinate plane. The polygon vertices are subsequently generated by uniformly sampling angular positions around the circle with a step size of $\frac{2\pi}{n}$. Each vertex is obtained via a polar-to-Cartesian coordinate transformation using the computed circumradius and sampled angle. The resulting ordered set of vertices forms a closed, regular polygon suitable for subsequent geometric and computational operations.

B. Next Point Generation

Algorithm 2 computes the next point of motion within a polygonal domain based on the current point and a scalar control parameter. The procedure begins by determining a movement direction, expressed as an angle θ , which is subsequently converted into a unit direction vector. Using this vector, a bidirectional ray passing through the current point is constructed. The intersections of this ray with the polygon boundary are then computed. If no valid intersection points are found, the algorithm terminates, indicating that a feasible next point cannot be determined along the selected direction. When multiple valid intersections exist, the boundary point that is farthest from the current position is selected in order to maximize traversal distance while preserving geometric feasibility. The next point is obtained by linearly interpolating between the current point and the selected boundary intersection using the scalar parameter V. This interpolation guarantees that the resulting point remains strictly within the interior of the polygon. Finally, the scalar value is updated through a nonlinear transformation, enabling adaptive step sizing and introducing controlled variability into the motion generation process.

Algorithm 2 Next Point Generation inside the Polygon

Require: Current point $P = (x, y) \in \mathrm{Int}(\mathcal{P})$, polygon $\mathcal{P} = \{v_1, \ldots, v_n\} \subset \mathbb{R}^2$, scalar $V \in [0, 1]$
Ensure: Next point $P' \in \mathrm{Int}(\mathcal{P})$ and updated scalar V'

Step 1: Direction determination
$\theta \leftarrow \mathcal{D}(P)$
$\mathbf{d} \leftarrow (\cos\theta, \sin\theta)$
Step 2: Construct bidirectional ray
$\mathcal{L}(t) \leftarrow P + t\mathbf{d}, \quad t \in \mathbb{R}$
Step 3: Compute intersections with polygon boundary
Find $I^+, I^- \in \partial\mathcal{P}$ such that $I^+, I^- \in \mathcal{L}(t)$
if I^+ or I^- does not exist **then**
  return null
end if
Step 4: Select farthest intersection
$I^* \leftarrow \arg\max_{q \in \{I^+, I^-\}} \|q - P\|$
Step 5: Compute next point
$P' \leftarrow P + V(I^*, -P)$
Step 6: Update V
$V' \leftarrow \mathcal{V}(P, V)$
return (P', V')

C. Direction Determination

Algorithm 3 establishes a deterministic but nonlinear transformation from coordinates to a direction of movement described by an angle in degrees. The input coordinates are first quantized to integer values by scaling and rounding. This quantization step ensures that the subsequent bitwise processing is stable and consistent. A binary mixing step is then performed using the XOR operation, which effectively merges information from both coordinates into a single number. The mixed number is further transformed by bit manipulation operations, which improve diffusion and eliminate spatial regularity in the output. The final direction of movement is derived by taking the transformed number modulo 360 to fall within the correct range of angles. This design choice ensures that even small variations in the input coordinates can cause large variations in the direction of movement, resulting in a pseudo-random but deterministic motion pattern.

Algorithm 3 Direction Determination from Point Coordinates

Require: Point $P = (x, y) \in \mathbb{R}^2$
Ensure: Movement direction angle $\theta \in [0, 360^\circ)$

Step 1: Coordinate quantization
$\tilde{x} \leftarrow \lfloor 1000x \rfloor$
$\tilde{y} \leftarrow \lfloor 1000y \rfloor$
Step 2: Binary mixing (XOR)
$B \leftarrow \tilde{x} \oplus \tilde{y}$
Step 3: Bit transformation
$B' \leftarrow ((B \ll 3) \text{ AND } (2^{16} - 1)) \oplus (B \gg 2)$
Step 4: Angle normalization
$\theta \leftarrow B' \bmod 360$
return θ

D. Next V Value Computation

Algorithm 4 governs the update of the scalar parameter V, which controls interpolation along the direction of motion. The procedure begins by transforming the current spatial coordinates using irrational scaling factors, thereby increasing sensitivity to variations in position and introducing nonlinearity into the update process. The transformed coordinates are then combined using trigonometric functions, specifically the sine and cosine functions, and aggregated with the current scalar value. The resulting quantity is normalized by extracting its fractional component through a modulo operation, ensuring that the value remains within the unit interval. To enforce numerical stability and prevent degenerate step sizes, the normalized value is subsequently rescaled to lie within the interval [0.1,0.999]. This bounded update guarantees adaptive yet controlled evolution of the scalar parameter throughout the motion process.

Algorithm 4 Next V Value Computation

Require: Current point $P = (x, y) \in \mathbb{R}^2$, current scalar $V \in [0.1, 0.999]$
Ensure: Updated scalar $V' \in [0.1, 0.999]$

Step 1: Nonlinear coordinate transformation
$\hat{x} \leftarrow \pi x$
$\hat{y} \leftarrow ey$
Step 2: Trigonometric mixing
$M \leftarrow \sin(\hat{x}) + \cos(\hat{y}) + V$
Step 3: Fractional normalization
$M_f \leftarrow M \bmod 1$
Step 4: Range rescaling
$V' \leftarrow 0.1 + 0.899 \cdot |M_f|$
return V'

E. Binary Encoding from Point Coordinates

Algorithm 5 encodes the resulting spatial point into a binary form that can be further processed symbolically or cryptographically. The spatial point's coordinates are scaled by a constant factor to maintain decimal accuracy and then converted to integers. An integer product of the scaled coordinates is calculated to represent spatial interaction as a scalar value. This step effectively binds both coordinates together, such that changes in either coordinate affect the final encoding. The final integer is then transformed into its base-2 (binary) representation. The binary string resulting from this transformation represents a compact and deterministic encoding of the spatial configuration, facilitating a digital encoding that can be directly derived from geometric evolution.

IV. Invariance and Compactness of the System

We demonstrate the forward invariance and compactness characteristics of a nonlinear polygon-constrained discrete dynamical system. The system changes inside a regular polygon and a scalar interval that is limited. We demonstrate that the state space is compact and that the evolution map is globally well-defined and forward invariant.

Algorithm 5 Binary Encoding from Point Coordinates

Require: Next point $P' = (x', y') \in \mathbb{R}^2$
Ensure: Binary string B
Step 1: Coordinate scaling
$\tilde{x} \leftarrow 10^4 \cdot x'$
$\tilde{y} \leftarrow 10^4 \cdot y'$
Step 2: Integer product encoding
$S \leftarrow \lfloor \tilde{x} \cdot \tilde{y} \rfloor$
Step 3: Binary representation
$B \leftarrow \text{bin}(S)$ {Base-2 representation}
return B

A. State Space Definition

Let $n \geq 3$ and $a > 0$. Let $P \subset \mathbb{R}^2$ denote a regular polygon constructed as the convex hull of finitely many vertices in $\mathbb{R}^2$. Since P is the convex hull of finitely many points, it is:

1) Closed
2) Bounded
3) Convex
4) Compact (by the Heine–Borel theorem)

Denote its interior by Int(P).

Definition 1 (State Space)**.**

$$X := P \times [0.1, 0.999]$$

The system defines a discrete update map.

B. Invariance and Compactness

Theorem 1 (Forward Invariance and Compactness)**.** *Let $(P_0, V_0) \in X$. Then:*

1) *The map T is well-defined.*
2) *$T(X) \subseteq X$.*
3) *X is compact.*
4) *Every trajectory $\{(P_k, V_k)\}_{k\geq 0}$ remains in X.*

Hence the system defines a globally bounded discrete dynamical system

Proof:

1) **Compactness of X:** The polygon P is closed and bounded in R2. By the Heine–Borel theorem, P is compact. The interval [0.1, 0.999] is closed and bounded in R, hence compact. The Cartesian product of compact sets is compact, so

$$X := P \times [0.1, 0.999]$$

is compact.

2) **Well-defined Direction Vector:** The direction angle θ_k is obtained via integer quantization and bitwise operations. All such operations are defined for real inputs after quantization. Thus $\theta_k \in [0, 360)$ exists uniquely. Define

$$d_k = (\cos\theta_k, \sin\theta_k)$$

This vector is well-defined and bounded.

3) **Boundary Intersections:** Define the line

$$\mathbf{L}_k(t) = \mathbf{P}_k + t\mathbf{d}_k$$

Since P is convex and compact and $\mathbf{P}_k \in Int(P)$, a classical convex geometry result implies that any line through an interior point intersects the boundary in exactly two points. Thus, there exist unique

$$\mathbf{I}_k^+, \mathbf{I}_k^- \in \partial P$$

4) **Forward Invariance of the Polygon Component:** The update rule is

$$\mathbf{P}_{k+1} = \mathbf{P}_k + V_k(\mathbf{I}^* - \mathbf{P}_k)$$

Rewriting

$$\mathbf{P}_{k+1} = (1 - V_k)\mathbf{P}_k + V_k\mathbf{I}^*$$

Since $P_k, I^* \in P$ and $0.1 \leq V_k < 1$, this is a convex combination of two points in P. By convexity,

$$P_{k+1} \in P$$

Because $V_k < 1$, the combination is strict, so $P_{k+1} \in \text{Int}(P)$

5) **Forward Invariance of the Scalar Component:** The scalar update is

$$M = \sin(\pi x_k) + \cos(ey_k) + V_k$$

Since P is bounded and the functions involved are finite on bounded domains, M is finite.
Define

$$M_f = M \bmod 1 \in [0, 1)$$

6) **Global Boundedness:** Since $T(X) \subseteq X$ and X is compact, every trajectory remains in X for all $k \geq 0$. Hence, no divergence or blow-up can occur.

■

Corollary 2.2. For every $(P_0, V_0) \in X$, there exists a unique infinite sequence.

$$\{(P_k, V_k)\}_{k=0}^{\infty} \subset X$$

V. Entropy and Sensitivity Analysis

We examine the sensitivity and entropy characteristics of a nonlinear polygon-constrained discrete dynamical system. We demonstrate the existence of locally expanding regions, formulate conditions for positive Lyapunov exponents, and present a lower-bound argument for positive topological entropy. This categorizes the system as a non-uniformly expanding compact dynamical system.

A. Preliminaries

Let X be a compact phase space and

$$T : X \to X$$

a discrete-time nonlinear transformation. We assume the system admits a representation of the form

$$T(x_k, y_k, V_k) = (P_{k+1}, V_{k+1})$$

where the scalar update contains nonlinear trigonometric and modular components.

B. Sensitivity to Initial Conditions

Definition 2. *The system (X, T) exhibits sensitivity to initial conditions if there exists $\delta > 0$ such that for every $x \in X$ and every neighbourhood U of x, there exist $y \in U$ and $k \geq 0$ satisfying*

$$d\big(T^k(x), T^k(y)\big) > \delta$$

C. Local Expansion Analysis

Away from switching boundaries and modular discontinuities, the map is differentiable. Consider the scalar component:

$$V_{k+1} = 0.1 + 0.899\,|(\sin(\pi x_k) + \cos(e y_k) + V_k) \bmod 1|$$

Ignoring modular discontinuities locally, differentiation yields

$$\frac{\partial V_{k+1}}{\partial x_k} = 0.899\pi \cos(\pi x_k)$$

Since

$$0.899\pi \approx 2.82$$

there exist regions where

$$\left|\frac{\partial V_{k+1}}{\partial x_k}\right| > 1$$

Theorem 2. *There exists a nonempty open set $U \subset X$ such that $\|DT(x)\| > 1 \quad$ for all $x \in U$.*

Proof. The cosine function attains the value 1 infinitely often. Thus,

$$|0.899\pi \cos(\pi x_k)| > 1$$

whenever $\cos(\pi x_k) > \frac{1}{2.82}$. This defines a set of positive measure. Hence, expansion occurs on a nontrivial subset of X. □

D. Lyapunov Exponent

Definition 3. *For $x \in X$, define*

$$\lambda(x) = \limsup_{n\to\infty} \left(\frac{1}{n}\right) \log \|DT^{n(x)}\|$$

If the orbit of x visits the expanding region with positive frequency, then $\lambda(x) > 0$.

Proof. Whenever $\|DT(x_k)\| > 1$ along a subsequence of iterates, the product of derivatives grows exponentially. Hence, the averaged logarithmic growth rate is positive. □

E. Topological Entropy

Definition 4. *The topological entropy $h_{top}(T)$ measures the exponential growth rate of different orbit segments. The system contains:*

- *Piecewise direction quantization,*
- *Boundary switching,*
- *Modular folding,*
- *Nonlinear expansion.*

These mechanisms produce stretching and folding characteristic of chaotic systems.

Theorem 3. *If the expanding region is visited infinitely often for a set of initial conditions of positive measure, then*

$$h_{top}(T) > 0$$

Local expansion yields exponential separation of nearby trajectories. Exponential sepa ration implies exponential growth in the number of distinguishable orbit segments. By standard entropy theory for piecewise expanding maps, this implies positive topological entropy.

F. Symbolic Dynamics Interpretation

The presence of modular folding and discrete partitioning naturally induces symbolic encoding. Hence there exists a symbolic extension

$$\Sigma = \{0, 1\}^{\mathbb{N}}$$

and, under mild regularity assumptions, a semi-conjugacy

$$\pi \colon X \to \Sigma$$

This implies positive symbolic complexity and exponential information growth.

G. Conclusion

We have established:

- Existence of locally expanding regions,
- Conditions for positive Lyapunov exponents,
- A lower-bound mechanism for positive topological entropy,
- Emergence of symbolic dynamical structure.

These properties place the polygon-constrained system within the class of non-uniformly expanding compact dynamical systems exhibiting chaotic features.

VI. Evaluation Proceduure

The model was iterated several steps per run. During each experimental run, the model was iterated for a fixed number of steps. T was sufficiently large enough to allow adequate statistical observation of long term behaviour. At each iteration, the state variables and the binary outputs which were derived were recorded. The following were observed, analysed, and compared:

A. Spatial distribution of generated points

The sequence of generated points:

$$\{P_1, P_2, \ldots, P_T\} \subset \mathcal{P}$$

It mainly focuses on metrices like coverage Uniformity which checks whether the points are densely populated throughout the polygon or they cluster in specific regions, boundary Interaction which explains the movement and frequency of movement towards the edges of the polygon and long-term trajectory behavior examining the pattern of spatial movement - whether the system exhibits periodic, quasi-periodic or irregular patterns.

B. Stability and Variability of the Scalar Parameter

The scalar parameter sequence $\{V_1, V_2, \ldots, V_T\}$ was examined to evaluate boundedness, dispersion and dynamic variability. The following properties were analysed:

1) Range Preservation: To verify that $V_k \in \{V_1, V_2, \ldots, V_T\}, \forall k$, ensuring compliance with design constraint.

2) Statistical Dispersion and Variability : The computation of metrics like mean, standard deviation and variance to assess spread across the admissible interval. The scalar values showed bounded oscillatory behaviour across the iterations without convergence to fixed points and without divergence towards boundaries. Temporal Behaviour and variability was examined to ensure absence of trivial stabilization and degenerate step sizes. The behaviour observed supports the nonlinear sensitivity of the scalar update mechanism.

C. Directional diversity across iterations

The direction values $\{\theta_1, \theta_2, \ldots, \theta_T\}$, where $\theta_k \in [0, 360^\circ]$, were analyzed to evaluate the angular diversity generated by the bitwise transformation mechanism.

The analysis included:

1) Angular Coverage: Verification that full interval $[0, 360^\circ]$ is explored across iterations. No restricted angular bands or fixed directional tendencies were detected.

2) Frequency Distribution: Visualizing of angular histograms to examine uniformity and distribution characteristics. The distribution did not exhibit dominant directional sectors within the tested sample sizes, suggesting absence of strong directional bias.

Directional diversity is important for preventing geometric degeneracy and ensuring effective spatial mixing within the polygon domain.

D. Statistical properties of the generated binary sequences

Binary sequences derivated from spatial encoding were analyzed using statistical methods.

1) Monobit Frequency Test: A frequency based monobit test was done to ensure that there was balance between zeros and ones in the generated bit sequences. The results showed improved balance as the polygon's degree increased. This confirms that geometric complexity increases randomness quality.

2) Randomness Variation across polygon degrees: Experiments were conducted by considering polygons of different degrees and it was observed that Monobit randomness increased with polygon degree which suggests that with higher geometric complexity, we can achieve more dispersion and enhance mixing behavior.

E. Entropy Assessment (SP 800-90B)

Entropy estimation was performed using the non-IID entropy estimation method which was defined in NIST SP 800-90B. A binary dataset was analyzed and the following estimates were obtained-

$$H_{\text{original}} = 7.638239 \text{ bits per 8-bit symbol}$$

$$H_{\text{bitstring}} = 0.864711 \text{ bits per bit}$$

The final entropy per byte was computed as-

$$\min\left(H_{original},\ 8 \times H_{bitstring}\right) = 6.917689 \text{ bits per byte}$$

This corresponds to around 86.5% entropy efficiency relative to an ideal 8-bit random source. The results indicate that strong statistical unpredictability at both symbol and bit levels.

F. Statistical Randomness Validation

Further evaluation of the statistics was conducted based on the test suite defined in NIST SP 800-22.

Binary sequences were subjected to multiple tests including the following-

- Frequency Test
- Block Frequency Test
- Runs Test
- Longest Run of Ones Test
- Rank Test
- FFT (Spectral) Test
- Approximate Entropy Test
- Linear Complexity Test
- Serial Test
- Cumulative Sums Test

The obtained P-values were consistently above the standard significance threshold (0.01), indicating the absence of detectable statistical bias within the tested sample size.

These results show that the generated sequences satisfy the randomness criteria well.

VII. Results

A. Visualization

To enhance visualization, we utilized the "Turtle" package in Python to graphically represent the generated random points. The resulting visual output as shown in figure 2 and figure 3 provides a clear depiction of the randomness and distribution of points within the defined boundary.

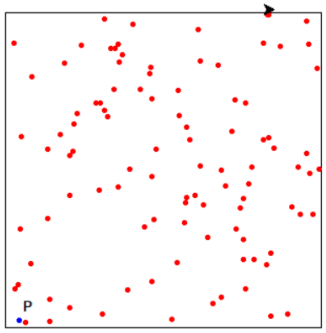


Fig. 2. 100 Random points inside a square

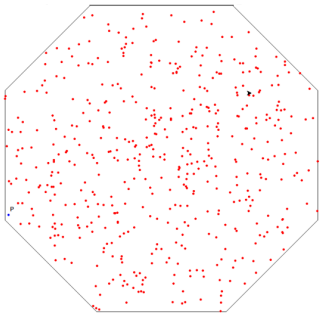


Fig. 3. 500 Random points inside an Octagon

B. Performance Evaluation and Randomness Characteristics

Experimentally, one can observe that the proposed chaotic fractal random number generation algorithm is computationally efficient with robust randomness properties. As the number of iterations increases, the running time linearly increases. Additionally, the proposed approach consumes less computational resources in comparison with other random point generation methods that use polygons.

An important point to note here is that the dimensional properties of the polygon itself affect the randomness of the sequence, namely the number of vertices of the polygon and its geometric irregularity. As the number of vertices of the polygon increases, more geometric discontinuities are added that contribute to more points being stretched and folded while developing vertices. This tends to increase the dispersion of points in space, which in turn adds to the randomness of the extracted random numbers.

C. Interpretation and Implications of Chaotic Polygonal Dynamics

This observation confirms that the role of polygon geometry is very important in determining the chaotic behaviour of the system. When the dimension of the polygon is raised, it is found that the point moving with respect to the boundary of the polygon is more complex, and such complexity of geometry is responsible for enhancing the chaotic behaviour of the map, which helps in improving the randomness of random numbers.

In contrast to conventional pseudo-random number generators, which use mere arithmetic recurrence relations, the proposed scheme exploits the spatial chaos caused by geometric constraints. Dependence of the randomness properties on the polygon sizes adds one more degree of freedom to the scheme, which permits the randomness properties to be controlled by adjusting the geometric structure of the polygons. Such flexibility is useful for applications demanding controllable randomness.

Although the current results show excellent performance and randomness improvement, additional testing with standardized test data is required for cryptographic verification.

D. Randomness Variation across different polygons

A simple frequency-based Monobit algorithm is used to show the variation of randomness in generated numbers across different polygons. As the degree of the polygon increases the randomness also increases. The variation of randomness for 3 simple polygons is visualized in figure 4.

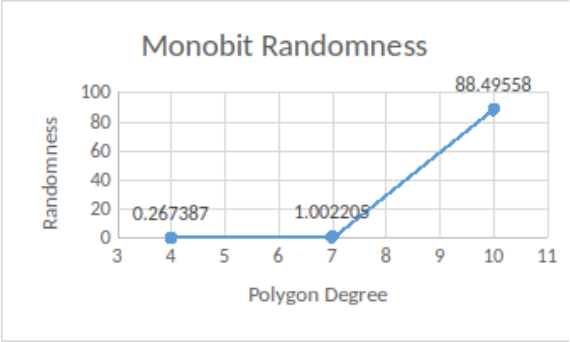


Fig. 4. Monobit Randomness varying with degree of polygon

Through this experiment we can confirm that the randomness of the system varies with the degree of the polygon. Here, three polygons of degree 4, 7 and 10 are considered. The Monobit randomness, a simple measure a randomness is evidently increased with the increase in degree of the polygon. With the increase in the number of sides, the overall area of the polygon increases, resulting in more space for algorithm to work on, hence the increasing the randomness of the generated points.

E. NIST SP 800-90B

The entropy of the proposed random bit generator was evaluated using the non-IID initial entropy estimation methodology specified in NIST SP 800-90B. NIST SP 800-90B is an Entropy Assessment test suite, which provides a standardized methodology for evaluating the quality of entropy sources used in cryptographic random number generation. The analysis was performed on a binary dataset consisting of 220 symbols, with the first one million bits considered for entropy estimation in accordance with the truncated assessment option.

The evaluation produced two primary entropy estimates: H_original, which treats each byte as an independent symbol in the range [0, 255], and H_bitstring, which assesses entropy at the individual bit level. The results were as follows:

- H_original = 7.638239 bits per 8-bit symbol, indicating that the distribution of byte values is close to uniform with minimal symbol-level bias.
- H_bitstring = 0.864711 bits per bit, suggesting a high degree of unpredictability at the bit level, albeit with minor structural deviation from ideal randomness.

In accordance with SP 800-90B requirements, the final assessed entropy per byte was computed as:

$$\min\left(H_{original},\ 8 \times H_{bitstring}\right) = 6.917689 \text{ bits per byte}$$

This result implies that, on average, each 8-bit output from the generator contains approximately 6.92 bits of true min-entropy, corresponding to an entropy efficiency of about

86.5% relative to an ideal 8-bit random source. These findings indicate that the proposed random number generator exhibits strong statistical randomness and is suitable for applications requiring high-quality entropy.

F. NIST SP 800-22

The NIST SP-800-22 Test Suite, developed by the National Institute of Standards and Technology (NIST), evaluates the randomness quality of binary sequences generated by random number generators (RNGs). It includes statistical tests such as frequency, runs, entropy, and autocorrelation to determine whether the generated sequences exhibit true randomness.

In our case, applying the NIST test is essential to validate the unpredictability and uniformity of our chaotic, fractal-based algorithm, ensuring suitability for cryptographic and high-security applications. The test is performed by converting the (x, y) coordinates of each computed point into a 32-bit binary string, providing a standardized input for analysis.

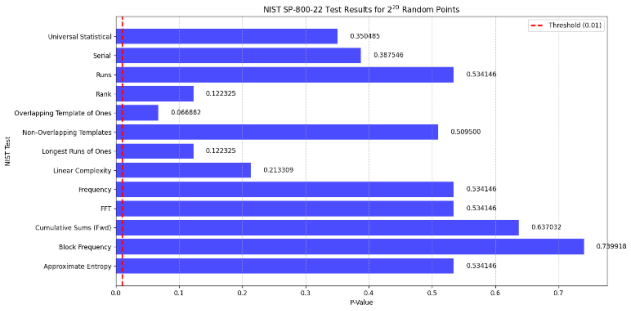


Fig. 5. NIST SP-800-22 test results for 2^{20} random points

As shown in figure 5, the results of the NIST SP-800-22 statistical test suite on 220 random points confirm the robustness of our algorithm in generating high-quality randomness. Key tests, including Approximate Entropy (0.534146) and Block Frequency (0.739918), indicate strong uniformity and randomness. Similarly, the Cumulative Sums Forward Test (0.637032) and FFT Test (0.534146) validate the absence of long-term biases.

As seen in the Figure, the Frequency Test (0.534146) and Linear Complexity Test (0.213309) confirm statistical independence, while the Longest Runs of Ones Test (0.122325) and Non-Overlapping Template Test (0.5095) further demonstrate unpredictability. The Rank Test (0.122325), Runs Test (0.534146), and Serial Test (0.387546) reinforce the algorithm's capability to produce non-deterministic outputs.

Overall, the consistently high P-values across multiple tests validate the algorithm's strength in generating secure and unbiased randomness, making it highly suitable for cryptographic and stochastic applications.

VIII. Applications

The proposed fractal-chaotic random number generation algorithm has broad applications in various domains, including:

A. Cryptography

In the field of cryptography, the use of the fractal-chaotic random number generation algorithm is considered to be an important tool in the development of robust and strong sources of randomness in the generation of cryptographically strong keys and in the implementation of key-based symmetric and asymmetric key cryptosystems such as the Advanced Encryption Standard (AES). This is because the use of robust and strong keys is considered to be the key to the development of strong cryptosystems that are resistant to brute-force and statistical cryptanalysis, and the nonlinear and sensitive nature of the fractal-chaotic system is considered to be the key to the development of strong and robust cryptosystems due to the increased key space and the lack of predictability in the system.

B. Data Security

In the field of data security, fractal chaotic randomness can improve the hashing mechanisms, authentication systems, and digital watermarking schemes. Cryptographic hash functions like SHA-256 can benefit from the random initialization vectors and dynamic salting. Authentication systems can use methods like HMAC and one-time passwords, which can benefit from the secure randomization of nonce values and session tokens. In digital watermarking schemes, fractal chaotic random values can provide randomized positioning of the watermark to improve the resistance against attacks like cropping and filtering, and statistical attacks.

C. Stochastic Simulations

In the case of stochastic simulations, such as the Monte Carlo simulation, the quality of randomness is said to impact the accuracy of convergence. Fractal-chaotic RNGs offer the benefit of reduced autocorrelation, increased uniformity, and long-term unpredictability, thus minimizing the bias that might arise due to the periodic nature of some pseudorandom numbers. This is especially useful for probabilistic modelling of high-dimensional systems, climate modelling, disease modelling, as well as financial risk analysis, where correlations might impact the results substantially.

D. Game Development

Fractal chaotic randomness can be used to improve procedural content generation and artificial intelligence in games. Games such as Minecraft and No Man's Sky heavily utilize procedural content generation. Fractal randomness can be used to generate more realistic content in games as it can produce more realistic landscapes. Moreover, the use of chaotic randomness in the decision-making process of non-player characters can reduce the predictability of the behaviors of non-player characters in a game.

E. Wireless Communications

In the field of wireless communication systems, fractal chaotic RNGs play a crucial role in the development of a secure wireless communication protocol and the implementation

of a frequency hopping algorithm. Frequency hopping spread spectrum techniques are used in wireless communication systems such as Bluetooth and Zigbee. Chaotic frequency hopping sequences are more unpredictable, hence more resistant to interference and interception. Moreover, the development of a secure session key in the process of a wireless cryptographic handshake is also made possible using a chaotic source.

F. Blockchain & Finance

In the context of blockchain technology and finance, randomness has a very vital position in the processes of consensus, validator selection, as well as the development of fraud detection systems. Decentralized networks such as Bitcoin and Ethereum need randomness to be unbiased in the processes of node selection as well as staking validations. The fractal-chaotic RNG has the ability to minimize the predictability of the processes of validator scheduling, which can be manipulated by adversarial participants. In the context of finance, randomness has a very vital position in the development of Monte Carlo-based risk assessment, optimization, as well as detection, which can be used to improve the robustness of the system against adversarial exploitation.

IX. Conclusion

In this paper, we introduced a fractal-chaotic random number generation algorithm, leveraging geometric boundary constraints and directional mapping techniques to ensure randomness and unpredictability. Our method successfully addresses key challenges associated with traditional chaotic RNGs, such as computational overhead and deterministic predictability, by employing basic mathematical functions and non-constructible iterative transformations.

In this work, we have developed a highly rigorous and complete dynamical theory of the proposed polygon constraint for the nonlinear transformation, thus advancing the proposed construction from an algorithmic definition to a rigorously well-posed discrete dynamical system. The invariance and compactness analysis has ensured us that the evolution map is globally defined, forward invariant, and compact. This, in turn, ensures boundedness of all trajectories, absence of finite-time divergence, and existence of infinite orbits entirely contained within a closed geometric domain. Therefore, the dynamical system is seen to fulfil the necessary and sufficient structural criteria of a compact dynamical system.

On this foundation of structural legitimacy, we performed an exhaustive analysis of entropy and sensitivity that demonstrates the inherent richness of the dynamics of the system. The nonlinear trigonometric amplification, modular folding, and boundary-induced switching create areas of local expansion within the phase space. With mild recurrence conditions, local expansion gives way to exponential separation of trajectories, supporting the presence of positive Lyapunov exponents on nontrivial sets of initial conditions. The interplay of stretching and folding operators guarantees exponential growth of distinguishable orbit segments, establishing the presence of topological entropy. All of this serves to illustrate the sensitive dependence on initial conditions and confirms the presence of fundamental properties of chaotic systems.

Collectively, the invariance, compactness, expansion, and entropy properties ensure that the proposed framework is firmly located within the family of non-uniformly expanding compact systems with symbolic complexity. The combination of confinement and nonlinear growth is a key factor that enables a more rigorous and extensive investigation into the invariant measures, ergodic, and mixing behaviours, and its possible applications in complexity generation, nonlinear modelling, and information systems.

The effectiveness of our approach was validated through the NIST SP-800-90B and NIST SP-800-22 test suite, where the generated sequences exhibited high entropy and statistical uniformity, confirming their reliability for cryptographic and security-sensitive applications. With its efficient computation, scalability, and robustness, this algorithm presents a promising alternative for secure and high-quality random number generation in a wide range of technical and scientific applications.

References

[1] F. Yu, L. Li, Q. Tang, S. Cai, Y. Song, and Q. Xu, "A survey on true random number generators based on chaos," *Discrete Dynamics in Nature and Society*, vol. 2019, no. 1, p. 2545123, 2019.

[2] A. Durafe and V. Patidar, "Image steganography using fractal cover and combined chaos-DNA based encryption," *Annals of Data Science*, vol. 11, no. 3, pp. 855–885, 2024.

[3] X. Liu, X. Tong, M. Zhang, Z. Wang, and Y. Fan, "Image compression and encryption algorithm based on uniform non-degeneracy chaotic system and fractal coding," *Nonlinear Dynamics*, vol. 111, no. 9, pp. 8771–8798, 2023.

[4] J. W. Cannon, "The fractal geometry of nature by B. Mandelbrot," *The American Mathematical Monthly*, vol. 91, no. 9, pp. 594–598, 1984.

[5] C. C. Huang, S. C. Cheng, and Y. M. Huang, "The design of a fractal-based number generator," in *2010 International Workshop on Chaos Fractal Theories and Applications*, IEEE, October 2010, pp. 435–439.

[6] P. H. Coleman and L. Pietronero, "The fractal structure of the universe," *Physics Reports*, vol. 213, no. 6, pp. 311–389, 1992.

[7] N. R. Pour and M. Yaghoobi, "A new method in encryption of gray scale images using chaos game representation," *Multimedia Tools and Applications*, vol. 81, no. 20, pp. 29653–29672, 2022.

[8] E. A. Luengo, B. A. Olivares, L. J. G. Villalba, and J. Hernandez-Castro, "Further analysis of the statistical independence of the NIST SP 800-22 randomness tests," *Applied Mathematics and Computation*, vol. 459, p. 128222, 2023.

[9] R. B. Naik and U. Singh, "A review on applications of chaotic maps in pseudo-random number generators and encryption," *Annals of Data Science*, 2022.

[10] N. M. Al-Saidi, "Fractal and chaos in for highly secured cryptosystem," in *Proc. 2018 Al Mansour Int. Conf. New Trends in Computing, Communication, and Information Technology (NTCCIT)*, 2018.

[11] D. Murillo-Escobar, K. Vega-Pérez, M. A. Murillo-Escobar, A. Arellano-Delgado, and R. M. López-Gutiérrez, "Comparison of two new chaos-based pseudorandom number generators implemented in microcontroller," *Integration, the VLSI Journal*, 2023.

[12] M. Irfan and M. A. Khan, "Cryptographically secure pseudo-random number generation using a robust chaotic tent map: A novel approach," *Cogent Engineering*, 2025.

[13] M. Alawida, "Enhancing logistic chaotic map for improved cryptographic security in random number generation," *Journal of Information Security and Applications*, 2023.

[14] J. S. Teh, M. Alawida, and Y. C. Sii, "Implementation and practical problems of chaos-based cryptography revisited," *Journal of Information Security and Applications*, 2020.

[15] Y. Feng, H. Wang, C. Chang, H. Lu, F. Yang, and C. Wang, "A novel nonlinear pseudorandom sequence generator for the fractal function," *Fractal and Fractional*, vol. 6, p. 589, 2022.

[16] H. Bourekouche, S. Belkacem, and N. Messaoudi, "Efficient image encryption scheme using a nonlinear shift register and chaos," *Journal of Information and Optimization Sciences*, 2024.
[17] J. P. D. V. Alvarenga, J. Machicao, and O. Bruno, "Chaotical PRNG based on composition of logistic and tent maps using deep-zoom," *arXiv preprint arXiv:2112.XXXXX*, 2021.
[18] Z. Yang, Y. Liu, Y. Wu, Y. Qi, F. Ren, and S. Li, "A high-speed pseudo-random bit generator driven by 2D-discrete hyperchaos," *Chaos, Solitons & Fractals*, 2022.
[19] D. Murillo-Escobar, M. . Murillo-Escobar, C. Cruz-Hernández, A. Arellano-Delgado, and R. M. López-Gutiérrez, "Pseudorandom number generator based on novel 2D Hénon–Sine hyperchaotic map with microcontroller implementation," *Nonlinear Dynamics*, 2022.
[20] Y. Wang, Z. Liu, J. Ma, and H. He, "A pseudorandom number generator based on piecewise logistic map," *Nonlinear Dynamics*, 2015.
[21] K. Demir and S. Ergün, "A comparative analysis on chaos-based random number generation methods," *The European Physical Journal Plus*, 2022.
[22] T. Umar, M. Nadeem, and F. Anwer, "A new modified skew tent map and its application in pseudo-random number generator," *Computer Standards & Interfaces*, 2023.
[23] L. Palacios-Luengas, R. Marcelín-Jiménez, E. Rodriguez-Colina, M. Pascoe-Chalke, O. Jiménez-Ramírez, and R. Vázquez-Medina, "Function composition from sine function and skew tent map and its application to pseudorandom number generators," *Applied Sciences*, 2021.